

Contestant Number: \_\_\_\_\_

Time: \_\_\_\_\_

Rank: \_\_\_\_\_



# COMPUTER NETWORK TECHNOLOGY (300) **STATE 2021**

*TOTAL POINTS*

\_\_\_\_\_ (100 points)

**Test Time: 60 minutes**

**Choose the *best* answer to the following questions. Mark A if the statement is *true*. Mark B if the statement is *false*.**

1. A PDU is also known by this name when referencing the Data Link layer.
  - A. packet
  - B. data
  - C. bit
  - D. frame
2. All of the following are disadvantages of peer-to-peer networks *except* they\_\_\_\_\_.
  - A. lack scalability
  - B. are not necessarily secure
  - C. are impractical for connecting large numbers of computers
  - D. centralize user account logins
3. Concerning twisted-pair cabling, at what maximum bend radius will data transmission most likely be disrupted?
  - A. No more than two times the diameter of the cable
  - B. No more than four times the diameter of the cable
  - C. No more than ten times the diameter of the cable
  - D. Twisted pair cable has a near infinite bend radius
4. \_\_\_\_\_is also known as a graphical representation of the infrastructure to a wired network.
  - A. network diagram
  - B. wiring schematic
  - C. topology map
  - D. system diagram
5. In VoIP, which device converts signals from analog phone equipment to IP data to travel over telephone lines?
  - A. VoIP PBX
  - B. VoIP MTA
  - C. VoIP gateway
  - D. VoIP translator
6. Which of the following is used to communicate with applications running on hosts at the Transport layer of the OSI model?
  - A. IP addresses
  - B. port numbers
  - C. domain names
  - D. MAC addresses

7. The command to display the configuration of TCP/IP for the network adapters installed.
  - A. ipconfig/show
  - B. ipconfig/list
  - C. ipconfig/all
  - D. ipconfig/full
8. The IPv6 address representing a global unicast address is \_\_\_\_\_.
  - A. FE80::10
  - B. 2000::/3
  - C. FC00::/7
  - D. FD00::/8
9. The tracert utility in Windows and the traceroute utility in Linux have similarities and differences. Which statement is *true* concerning one of their differences?
  - A. Only tracert can sent UDP messages for tracing a path.
  - B. By default, the tracert utility uses ICMP echo requests, while traceroute uses UDP datagrams or TCP SYN messages.
  - C. The Windows tracert utility does not place limits on the TTL of repeated trial messages.
  - D. The tracert utility expects an ICMP port unreachable error message as the final reply to a trace.
10. IP does *not* establish a session to send packets.
  - A. True
  - B. False
11. What is the result of a router receiving a packet with a TTL equal to 0?
  - A. The router drops the packet and sends an ICMP TTL expired message back to the host.
  - B. The router attempts to forward the traffic on a local network.
  - C. The router resets the TTL to 128.
  - D. The router marks the packet as corrupted and forwards it to the next hop.
12. Which unit is employed to measure the distortion of a signal on a network?
  - A. joules
  - B. volts
  - C. decibels
  - D. farads

13. Which twisted-pair cabling standard must be utilized for wiring a new government building?
- A. TIA/EIA 568A
  - B. TIA/EIA 568B
  - C. TIA/EIA 568F
  - D. TIA/EIA rollover
14. Which of the following Ethernet cabling standard employs the fastest speed?
- A. 100Base-T
  - B. 1000Base-T
  - C. 10GBase-T
  - D. 10TBase-T
15. When examining 802.11, at which layer of the OSI model might you see the standards vary.
- A. Physical Layer
  - B. Network Layer
  - C. Data link Layer
  - D. Transport Layer
16. When using a Linux device to connect to an SSH server, what location must house the public key before the device is allowed to connect?
- A. In an authorization file under your home directory on your computer
  - B. In an authorization file on the host where the SSH server is
  - C. In the /etc/ssh/keys folder
  - D. In the /var/run/ssh/public folder
17. The IPsec protocol operates at this layer of the OSI model.
- A. Physical Layer
  - B. Network Layer
  - C. Transport Layer
  - D. Application Layer
18. Two different, native VLAN connected access ports that both tag traffic with different VLAN IDs will trigger a VLAN mismatch.
- A. True
  - B. False
19. The CIDR notation for a subnet of 255.255.248.0.
- A. /18
  - B. /20
  - C. /21
  - D. /29

20. Networks are typically segmented in all of the following ways *except* by \_\_\_\_\_.  
A. geographic location  
B. departmental boundaries  
C. device types  
D. device manufacturer
21. The words malicious and software combine to form the term malware.  
A. True  
B. False
22. Which definition explains the function of a worm?  
A. A program that disguises itself as something useful but actually harms your system.  
B. A process that runs automatically, without requiring a person to start or stop it.  
C. A program that runs independently of other software and travels between computers and across networks.  
D. A program that locks a user's data or computer system until a ransom is paid.
23. By definition, which is considered a phishing attempt?  
A. An employee at your company has received a malware-infected file in their e-mail.  
B. A person posing as an employee tried to access a secured area at your organization.  
C. A gift was offered to an employee with access to secured information in exchange for details.  
D. An e-mail was sent to a manager at your company that appeared to be from the company's CTO asking for access.
24. What is the equivalent wildcard mask value of using the "any" keyword when used in access control lists?  
A. 0.0.0.0  
B. 255.255.255.255  
C. 255.255.0.0  
D. 0.0.255.255
25. What is the first step in STP path selection?  
A. STP must first select the root bridge, or master bridge.  
B. STP examines the possible paths between all other bridges.  
C. STP disables links that are not part of a shortest path.  
D. STP begins to block BPDUs on non-designated ports.

26. A leased frame relay circuit holds this advantage.
- A. You are guaranteed to receive the maximum amount of bandwidth specified in the circuit contract.
  - B. You pay only for the bandwidth you require.
  - C. The paths that your data will take are always known.
  - D. Frame relay is a newly established network technology with more features than other technology.
27. After identifying the problem, a theory to a probable cause of a problem should be established.
- A. True
  - B. False
28. What type of failure is caused by an open electrical circuit in a failed circuit breaker?
- A. fail-open
  - B. fail-close
  - C. fail-tolerant
  - D. fail-dynamic
29. What is the Presentation layer responsible for in the OSI model?
- A. The Presentation layer describes the interface between two applications, each on separate computers.
  - B. The Presentation layer is responsible for reformatting, compressing, and/or encrypting data in a way that the application on the receiving end can read.
  - C. The Presentation layer is responsible for describing how data between application is synced and recovered if messages don't arrive intact at the receiving application.
  - D. The Presentation layer is responsible for transporting Application layer payloads from one application to another.
30. Which of the following is an example of encapsulation?
- A. The addition of a header to data inherited from the layer above in the OSI model
  - B. The subtraction of a header from data inherited from the layer below in the OSI model
  - C. The modification of headers from a higher layer in the OSI model
  - D. The addition of a trailer to data inherited from the layer above in the OSI model
31. Which layer of the OSI model do the TCP and UDP protocols operate?
- A. Application Layer
  - B. Presentation Layer
  - C. Transport Layer
  - D. Network Layer

32. In a rack, multiple consoles can connect to a single device using a KVM.
- A. True
  - B. False
33. A cable bend radius can be best defined as a \_\_\_\_\_.
- A. cable bend radius is the radius of the maximum arc into which you can loop the cable without impairing data
  - B. cable bend radius is the radius of the minimum arc into which you can loop the cable without impairing data
  - C. cable bend radius determines the total number of bends that can exist in an overall cable before data is impaired
  - D. cable bend radius defines the twist rate of the cable inside of the insulation
34. In a network diagram, which symbol represents a network switch?
- A. An icon that has a hockey-puck shape with two arrows pointing inward and two arrows pointing outward
  - B. An icon that resembles a brick wall
  - C. An icon that is rectangular in shape, with squiggly lines on the front
  - D. An icon that is rectangular, which contains four arrows pointing in opposite directions
35. TLD servers can be located at the root level of the DNS hierarchical structure.
- A. True
  - B. False
36. The NIC of a computer has an IP of 169.254.196.200. What can the technician assume from this information?
- A. The computer has been assigned a routed public IP address.
  - B. The network card has been erroneously assigned a loopback address.
  - C. The computer is configured to use DHCP, but was unable to lease an address.
  - D. The network card is set up for multicast communication.
37. All IP address ranges are recommended for use private networks except \_\_\_\_\_.
- A. 10.0.0.0 through 10.255.255.255
  - B. 172.16.0.0 through 172.31.255.255
  - C. 192.168.0.0 through 192.168.255.255
  - D. 127.0.0.0 through 127.255.255.255

38. During DHCP transmission, clients receive responses on which port?
- A. port 67
  - B. port 68
  - C. port 546
  - D. port 547
39. BGP is accurately defined by the following.
- A. BGP is limited to a single autonomous system.
  - B. BGP is exclusively a distance-vector protocol.
  - C. BGP utilizes TCP for communicating updates.
  - D. BGP is a more advanced version of OSPF.
40. If duplicate MAC addresses exist on a switched network, what will the outcome be?
- A. The hosts that share the same MAC addresses will be completely unable to communicate with any other devices.
  - B. The hosts will generate new MAC addresses until the conflict is resolved.
  - C. The hosts will still send and receive traffic, but traffic may not always reach the correct destination.
  - D. The network switch will eventually crash due to being unable to properly forward traffic.
41. Flaws in transmission from attenuation is most likely caused by \_\_\_\_\_.
- A. a customer modem is continuously losing signal due to large distance from the transmitting device
  - B. a network switch interface is dropping due to a powerful radio transmitter in the adjacent room
  - C. a customer's video conferencing application is experiencing delay in audio and video that comes and goes
  - D. two network interfaces are experiencing problems at the same time due to the cables having adjacent runs
42. How deep should the sheath of an Ethernet cable extend into the terminating RJ-45 connector?
- A. 1/2 inch
  - B. 3/8 inch
  - C. 4/5 inch
  - D. 1/5 inch
43. A shared access point, by a group of workstations, is called a(n) \_\_\_\_\_.
- A. extended service set
  - B. basic service set
  - C. modified service set
  - D. generic service set

44. What is the frequency range for the wireless spectrum, as defined by the FCC?
- A. 9 kHz and 300 GHz
  - B. 125 kHz and 150 GHz
  - C. 3 kHz and 500 GHz
  - D. 3 MHz and 300 GHz
45. The term used to define a service provider that rents space at a data center is \_\_\_\_\_.
- A. point of presence (PoP)
  - B. service location (SL)
  - C. central service point (CSP)
  - D. locally exchanged data point (ledp)
46. The amount of /64 subnets that can be created within /56 prefix is \_\_\_\_\_.
- A. 256
  - B. 512
  - C. 1024
  - D. 2048
47. Where is the VLAN database stored in VTP?
- A. On the router responsible for maintaining the VTP database
  - B. On the switch that is configured as the trunk root
  - C. On the switch that is known as the stack master
  - D. On the designated VLAN server for the network
48. Which legacy protocol requires both devices to authenticate?
- A. Password Authentication Protocol (PAP)
  - B. Challenge Handshake Authentication Protocol (CHAP)
  - C. Microsoft Challenge Handshake Authentication Protocol (MS-CHAP)
  - D. Microsoft Challenge Handshake Authentication Protocol, version 2 (MS-CHAPv2)
49. Which of the following statements best describes role-based access control?
- A. RBAC allows a network administrator to base privileges and permissions around a detailed description of a user's roles or jobs.
  - B. RBAC allows users to decide for themselves who has access to that user's resources.
  - C. RBAC organizes resources into hierarchical classifications, such as "confidential" or "top secret".
  - D. RBAC is the most restrictive method of access control.
50. Cable management is *not* important in a rack system.
- A. True
  - B. False